Defense Production and Sustainment Facility-Class Design Basis

The First Facility Class of the Defense Industrial Base Sector Design Basis

DBA-DIB-DP-FC1 — Facility-Class Design Basis, DBA-DIB Series

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document sets out the design basis for defense production and sustainment, the first facility class in the DBA-DIB series. The *Defense Industrial Base Sector Framework* (DBA-DIB_Sector_Framework) states the sector's design basis in general terms, and this document applies it to one kind of facility: the plant, yard, arsenal or depot that builds the materiel with which the force fights and keeps it in service. It names the events such a facility must withstand, traces how each event reaches the force and the nation, sets the parameters that size the class's protection, and states what must be true of a production capability, of the technical data it holds and of the output it sends to the force for the capability to be sustained and not handed to an adversary. It describes no particular plant. A design basis for a specific plant, program or yard starts from this document and adds only what is particular to it.

The framework sets two features apart in this sector. The first is that its governing threat is compromise as well as disruption, so that alongside the availability consequence every other producer sector carries, it carries a confidentiality-and-integrity consequence that is not recoverable: an exfiltrated design and an implant already in the supply chain are not undone by rebuilding a plant (framework §1.5). The second is that it is downstream of Critical Manufacturing, so that it inherits that sector's availability analysis and adds its own (framework §1.5, §6.1). It therefore requires two consequence measures and refuses to collapse the second into the first (framework §3.6). This document takes both up for the production function, which the framework says spans both axes (framework §3.2). On the author's decisions of 14 October 2026 the class is the sector's first and proposes three reference events of its own (Section 9.2).

1.2 Scope

The class covers the facilities the framework's register names for the function: weapons-systems production, aerospace manufacturing, shipyards and ammunition plants, together with the government-owned contractor-operated facilities, arsenals and depots that produce and sustain materiel (framework §3.2, §8.2). For a facility it covers the final assembly and integration of an end item, an aircraft, a missile, a vehicle, a ship or a munition; the loading, assembly and packing of munitions, with the energetic materials that make them hazardous; the depot-level maintenance, overhaul and repair that keep fielded systems in service; the technical data package from which the facility builds and sustains, which it holds and which an adversary seeks; the acceptance of each unit before it goes to the force; and the cleared and skilled workforce that does all of it. It includes the facility's surge, the rate it can reach in war above its rate in peace, because for this function the gap between the two is itself a design-basis quantity.

Three neighbors are separate classes of this series. The sole-source component suppliers, rare-earth materials and microelectronics fabrication on which production depends are *Defense Supply Chain* (DBA-DIB-SC-FC2). The research and development facilities, test ranges and test-

and-evaluation systems where capability is designed and proven are *Research, Development, Test and Evaluation* (DBA-DIB-RD-FC3). Acquisition, Defense Production Act mobilization and industrial-base policy are the coordination class, DBA-DIB-CorpCo. They are covered here only where the production facility depends on them. The design data a production facility holds in order to build is this class's; the design data a research facility holds in order to design is the research class's, and where one site holds both, both classes apply (Section 8.3).

Two boundaries with other series are drawn in the framework and taken here as given. The long-lead production equipment and the materials a defense plant depends on are Critical Manufacturing's, whose first class, *Long-Lead Critical Components* (DBA-CM-LL-FC1), states the manufacturing side of RT_HVLLC; this class applies that statement to defense production and does not restate it (framework §1.2, §6.3). And covert-influence operations against the industrial base fall under the surveillance sibling standard, not here (framework §6.3). Section 8.3 states the boundaries.

What makes a facility a member of this class is the same at every site: it builds or keeps in service materiel the force cannot fight without, from a technical data package an adversary would take if it could, and the output it sends to the force must be what it is said to be. The particular end item, process and data differ from site to site, and that detail belongs in each facility's own design basis.

1.3 The Reference Facilities

Three kinds of facility are used as reference cases. Each was chosen because it shows a different way the capability can be lost or compromised.

The ammunition and energetics plant loads, assembles and packs munitions, or makes the propellants and high explosives that fill them, often in a government-owned plant run by a contractor. It is the governing configuration for the availability axis, because its output is consumed in war at a rate far above what it makes in peace, because its energetic materials make it hazardous in a way no other class in the series is, and because some of its inputs have a single domestic source. The record of the effort to raise the production of 155 mm artillery shells after 2022 shows the configuration under demand (Section 5.4). The Ukrainian strikes on Russian defense production, conferred by the framework as DBA-DIB-RE-2 (framework §9.1), show such a plant as a target (Section 6.3). The fire at a Diehl group site in Berlin in May 2024, which this class proposed for sabotage, is conferred as DBA-DIB-RE-1 and is read under §5.1 as a loss by fire, on the author's ruling of 14 October 2026 to follow the criminal record, held as the RECORD PRIMACY convention of the DBA Reference Event Register (Section 5.1); the site, operated by Diehl Metal Applications, coats and stamps metal parts and, on the company's account, does not supply Diehl Defence, so it is not an instance of this configuration.

The systems-integration plant builds and integrates aircraft, missiles or other complex end items, and holds the technical data package of the programs it builds: the drawings, specifications, test data and manufacturing processes from which an adversary could learn the system or build a

copy. It is the governing configuration for the confidentiality axis. The theft of data on the C-17 transport aircraft and on fighter aircraft from defense contractors' networks between 2008 and 2014, conferred by the framework as DBA-DIB-RE-3 (framework §9.1), is the case (Section 5.5).

The shipyard and depot builds, overhauls and repairs ships, submarines, aircraft and ground systems. It is the reference for sustainment: for the facility whose loss does not stop the production of new materiel but takes fielded materiel out of service, and for the capability that exists at so few sites that a single loss removes a large share of it. It is used here as the reference for the dry dock, the heavy lift and the specialized test facility that are themselves long-lead.

1.4 Class Type

This is a physical facility class under the author's determination of 30 September 2026 (Facility-Class Types Are Physical, Coordination and Logical). The harm it governs runs through plants, yards, depots and their equipment, through the energetic materials some of them handle, and through the people who work there. In the framework's structure it sits under Defense Production and Sustainment (DP), the first of the sector's three function classes (framework §3.1, §3.2). It carries one thing no physical class before it has carried: the technical data the facility holds is an asset in its own right, whose loss is complete when it is copied and is not restored by anything the facility can rebuild. The class therefore states a confidentiality design basis for a physical facility, and it states it as the framework requires, on its own axis (framework §3.6).

1.5 Relationship to Governing Documents

This instrument is issued under the Defense Industrial Base Sector Framework and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, bounding principles, dual-axis consequence or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton (framework §4, §5 and §7), this instrument develops the class-level content ahead of it and records in Section 9 what the framework should take up. Section references of the form framework §N resolve against the Defense Industrial Base Sector Framework in force. RT_HVLLC is the energy series' metric, adopted by Critical Manufacturing as its native metric and inherited by this sector on its availability axis (framework §3.6); this class cites *Long-Lead Critical Components* for its manufacturing side and does not restate it.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the

authorization obligation may be discharged, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*Envelope*) reserves the term envelope. Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

2. The Defense Production and Sustainment Facility Class

2.1 Defining Characteristics

Six characteristics define the class, and each is a reason the loss or compromise of a facility reaches farther than the loss of most facilities.

The first is the limited source. For many end items and munitions there is one prime, one final-assembly line, one loading plant or one yard able to build or sustain them, and for some of their inputs, among them certain energetic materials, there is one domestic plant. The framework names sole-source and limited-supplier concentration as the sector's structural exposure (framework §2.1).

The second is the long qualification. A defense production line is qualified to the program it builds: its processes, its tooling and its people are proven against the technical data package, and a new line, or an old line restored after a loss, must be qualified again before its output is accepted. The time to qualify is part of the time to recover, and the record of 2022 to 2026 shows that a new line can take years to deliver acceptable output (Section 5.4).

The third is the surge. In peace the force buys at a rate that sustains training and stockpile; in war it consumes at a rate many times higher. A facility's design basis is not met by its peacetime rate, and the capacity to surge, the tooling, the energetics, the people and the time to reach the higher rate, must exist before it is needed.

The fourth is the technical data. The facility holds the data from which the end item is built: drawings, specifications, process instructions and test results, much of it export-controlled and some of it classified. That data is what an adversary seeks when it seeks to learn a system's weaknesses or to build its own, and its loss is complete when it is copied (framework §1.5).

The fifth is that the output enters the force. A round, a missile or an aircraft that leaves the facility is fielded and may be used years later in conditions no test reproduces. An output that is not what it is said to be, through a defect, a counterfeit part or a deliberate alteration of the process, fails when it is needed, and the failure may not be traced until many units are in service.

The sixth is the hazard. Facilities that make or load energetic materials can be destroyed by their own product, and an attack or an accident there can cost lives on site and the capability with them. The class carries the hazard of a chemical plant and the consequence of a sole-source producer at once.

2.2 The Classification and Sole-Source Sub-Index

The framework sub-indexes the sector's classes by security classification and sole-source criticality, so that the design-basis load a facility carries reflects both the sensitivity of what it holds and the irreplaceability of what it makes (framework §3.1, §8.2). The two dimensions are the two axes, and this instrument states three tiers on each. A facility's tier is stated on each axis separately and is not averaged.

On the availability axis, the first tier is the facility that is the sole or one of very few sources of an end item, a munition or a sustainment capability the force cannot fight without, or the sole domestic source of an input to many of them, and whose loss could not be made good within the time a conflict would last. The second tier is the facility whose capability is shared with others of the same kind, so that its loss lengthens supply without ending it. The third tier is every other facility in the class.

On the confidentiality-and-integrity axis, the first tier is the facility that holds the technical data package of a fielded or major program at a level whose loss would give an adversary a decisive advantage, or whose output is a system on which the force's survival in combat depends. The second tier is the facility that holds partial or export-controlled data of such a program, or makes components whose compromise could be found in acceptance. The third tier is every other facility in the class, which still holds controlled unclassified information and carries the obligations that come with it.

2.3 Deployment Typologies and Size Range

The class spans several kinds of facility. An ammunition plant receives metal parts, propellants and high explosives, and loads, assembles and packs them into rounds, often in buildings separated by distance against the chance that one explodes. An energetics plant makes the propellants and explosives themselves from chemical precursors. A systems-integration plant receives subassemblies from a supply chain that may number thousands of firms, integrates them and tests the end item, and holds its technical data on networks connected to its suppliers and to the government. A shipyard builds or overhauls hulls in dry docks and on building ways with heavy lift. A depot overhauls and repairs fielded systems. Some are owned by the government and run by a contractor; some are the government's own arsenals and yards; most are commercial plants under defense contracts.

The range is wide. At one end is a single plant that is the nation's only source of an energetic material on which most of its munitions depend; at the other a machine shop that makes a part for many programs and holds a fragment of each program's data. The class is written to fit both, because the consequence of either is measured in the capability of the force.

2.4 Consequence Context

The class's consequence runs on two axes, as the framework requires (framework §3.6). On the availability axis it is the capability the force loses when a facility cannot produce or sustain,

measured in time, the time to restore the production or sustainment rate that the force's design basis counts on. On the confidentiality-and-integrity axis it is the capability compromised when the facility's data is taken or its output corrupted, measured not in time but in scope, the programs, generations and fielded units a single compromise reaches. The framework adds that this second consequence flows outward as a gain to the adversary rather than only a loss to the defender (framework §6.2).

The record shows the availability axis under demand. Before Russia's full-scale invasion of Ukraine in 2022 the Army produced about 14,000 rounds of 155 mm artillery ammunition a month; the Department set a goal of 100,000 a month by October 2025; and in July 2026 its Inspector General reported that production stood at about 36,000 rounds a month as of March of that year, after $4.5 billion spent, and that a contractor-operated plant in Mesquite, Texas, built to make projectile metal parts, had not yet produced a single shell meeting the Army's specifications, as Stars and Stripes reported on 14 July 2026. And it shows the confidentiality axis exercised. Between 2008 and 2014, on the Department of Justice's account, a conspiracy directed by a businessman in Canada and carried out by two people in China, whom the Department described as hackers of the People's Liberation Army Air Force, took data on the C-17 transport aircraft and on certain fighter aircraft from defense contractors, including computers belonging to Boeing, and sent it to China (Section 5.5).

3. The Facility-Class Design Envelope

3.1 Why Defense Production Is the First Class

Three reasons put this class first. It is the function the framework names first, and the one in which both of the sector's distinguishing features meet: the framework states that its governing consequence spans both axes (framework §3.2). It is the class most directly tied to the series already written: its availability axis is the one *Long-Lead Critical Components* has stated for Critical Manufacturing, and it can be stated now by applying that class rather than waiting for the framework's Section 4. And it is the class whose events the plan for closing the seven sectors named: a fire at a defense plant in Europe, strikes on a belligerent's defense production, and the theft of a program's data from defense contractors. The supply-chain and research classes, whose consequence lies more on the confidentiality-and-integrity axis alone, are left to their own instruments (Section 9.2).

3.2 The Three Consequence Chains

Every event in the class reaches the force by one of three chains. The first is on the availability axis; the second and third are the two halves of the confidentiality-and-integrity axis, which the framework names together and this instrument separates because they are reached by different paths and bounded by different means.

Chain A, the availability chain, begins with something that removes or limits the capability to produce or sustain: the loss of a facility, of its line, of a critical input or of its people, or a

demand that outruns its rate. The force receives fewer rounds, aircraft or overhauled ships than its own design basis counts on, for as long as the capability takes to restore or to expand. Its consequence is measured in the time to restore the rate, and it is the chain the class shares with Critical Manufacturing.

Chain X, the exfiltration chain, begins with an adversary who reaches the technical data the facility holds and copies it. Nothing is lost at the facility, which may not know it has happened for years. The adversary learns the system, finds its weaknesses or builds its own, and the advantage the program was meant to give is reduced or reversed for every unit already fielded and every unit yet to be built. Its consequence is measured in scope, and it is complete at the moment of the copy (framework §3.6).

Chain I, the integrity chain, begins with something that makes the facility's output other than what it is said to be: a counterfeit or altered part accepted from the supply chain, a process changed without authority, or an acceptance test that does not detect the change. The output enters the force and fails, or can be made to fail, when it is used. Its consequence is measured in scope, the units affected, and it is not bounded until every affected unit has been found.

The three chains are joined by the facility's networks. The same connection through which a prime shares its technical data with its suppliers is the path an adversary uses to take it; the same control system through which a line is run is the path by which its process can be changed. And they are joined by the adversary's purpose: a campaign that takes a program's data in peace and strikes its production in war has opened all three.

3.3 Design Parameters at Class Resolution

Six parameters govern the envelope, three on each axis.

The first is the production rate and surge rate: for each end item, munition or sustainment capability a facility provides, its rate in peace, the rate the force's design basis requires in war, and the time and inputs needed to reach it. The gap between the two, and the time to close it, is the class's first availability quantity.

The second is the qualified source count: the number of facilities, and for each critical input the number of plants, able to provide the capability at qualified quality, with the share of them under domestic or allied control. It is the parameter *Long-Lead Critical Components* states at its §3.3, applied here to the defense end item and its inputs.

The third is the production recovery time: the time to restore a facility's rate after its loss, which is the RT_HVLLC of its long-lead production equipment, stated by *Long-Lead Critical Components* as queue time plus build time plus delivery and installation (*Long-Lead Critical Components* §3.3), plus the time to requalify the line and its output. It is how this class applies the framework's adoption of RT_HVLLC on the availability axis (framework §3.6).

The fourth is the data holding: for each program, every place its technical data package is held, by the prime, its suppliers and the government, at what level of control, on what networks and

with what connections. It sets how many paths Chain X has, and the framework's confidentiality measure cannot be applied without it. The holding includes its index and its people. In the C-17 case a co-conspirator sent the directory listings of what he had reached, and the man directing him chose from them which files to steal, so that the listing was the map of the theft; listings and metadata are therefore protected at the level of what they describe, their export is made detectable, and the compromise scope is assessed on the assumption that the index has been taken (*The Theft of C-17 and Fighter Aircraft Data From Defense Contractors* F-1, R-1). And the people with access, their roles and their exposure are part of the holding, and their targeting is an attack on it (*The Theft of C-17 and Fighter Aircraft Data From Defense Contractors* F-2, R-2; noted to *Defense Supply Chain*).

The fifth is the compromise scope: for each holding, the programs, system generations and fielded units whose design advantage would be lost if that holding were copied. It is this instrument's statement of the framework's non-recoverable measure, the scope of capability compromised rather than a duration (framework §3.6), and Section 10.2 records it for the framework to take up.

The sixth is the provenance coverage: the fraction of the parts, materials and process changes in a unit whose origin and authority the facility can state and verify at acceptance, and the fraction of fielded units for which it can state what they contain. It sets how far Chain I can run before it is found.

This instrument states the parameters and their governing logic. Their quantitative values for a facility or program await validation against what the government and industry already hold: the industrial-base assessments the Department prepares, the munitions requirements of the services, the contractors' own records of their data holdings and the assessments made under the cybersecurity requirements of Section 8.4. Much of that data is itself controlled or classified, which is why the values are not asserted here and why a facility's design basis that states them must be held at the level of what it states (Section 10.1). The figures quoted are their sources' own.

4. The Production Line and the Technical Data Package

4.1 Status and Use

Every event in this class reaches the force through one of two things the facility keeps: the production line, with its inputs, its rate and its acceptance of what it makes, and the technical data package from which it builds. They are the class's governing pathway, and this section states what the class requires of each. It is written as the receiving function of the hospital class, DBA-HPH-CD-FC1 (*Care Delivery*), and the order book of *Long-Lead Critical Components* are written, because the line and the data are the facility's interfaces with the force it supplies and with the adversary who would deny or exploit it.

4.2 The Surge as a Design-Basis Quantity

A facility's peacetime rate is usually regarded as a matter of the budget. In this class the war rate is a design-basis quantity, because the force's own design basis counts on it. The class requires that a first-tier facility's design basis state, for each capability it provides, its peacetime rate, the war rate the force's requirement sets, the time to reach the war rate from a standing start, and what reaching it depends on: tooling, energetics and other inputs, qualified people and the facilities, including new ones, that would add capacity. It requires that new capacity be credited only from the date its output is accepted, not from the date it is built. The record of the 155 mm effort shows why: a plant built to add capacity was, on the Inspector General's account, still not producing acceptable output in 2026 (Section 5.4). The Defense Production Act gives the government the authority to direct priority and to finance capacity (Section 8.4); the surge rate is what that authority can actually produce, and it is the coordination class's to plan.

4.3 Single Points in the Inputs

A line can run at its rate only if its inputs arrive at that rate. The class requires that a first-tier facility's design basis state, for each input without which it cannot produce, the number of qualified sources, the share under domestic or allied control, the stock held and the time that stock covers at the war rate. Where an input has a single domestic source, among them some energetic materials and some large castings and forgings, the source's loss is a design-basis event for every facility that depends on it (Section 5.3), and its resilience is stated in the class that governs it: the defense supply-chain class for defense-specific components, *Long-Lead Critical Components* for long-lead production equipment, and the chemical series for the precursors of energetics (Section 8.2). The class's own requirement is that the dependence be stated and that the stock or the second source be sized against the war rate.

4.4 The Data Holding as a Design-Basis Asset

The technical data package is usually regarded as a matter of security compliance. In this class it is a design-basis asset, because its loss is the class's non-recoverable consequence. The class requires that a first-tier facility's design basis state the data holding of Section 3.3 for each program: where the package is held, in whole or in part, by the facility, its suppliers and the government; on what networks; with what connections between them; and who can reach each holding. It requires that the package be divided so that no single holding, and no single intrusion, reaches more of it than the work at that holding needs, and that each holding's compromise scope be stated. It requires that the facility be able to detect the copying of a holding within a stated time and to state, after an intrusion, what was reached. Certification under the requirements of Section 8.4 is evidence toward these requirements and does not by itself meet them; the framework records that security compliance is measured in controls and a design basis in demonstrated capability (framework §2.3). The statement of the holding names its index and its people (Section 3.3).

The prime contractor's connection to its suppliers is part of the holding. A prime that shares its package with a thousand suppliers has a thousand holdings, and an adversary needs only the weakest. The class requires that the prime's design basis state the holdings at its suppliers for its first-tier programs and the scope each reaches, and refers the suppliers' own protection to the defense supply-chain class (Section 8.3).

4.5 Acceptance and Provenance

A unit is accepted before it goes to the force, and acceptance is where Chain I can be stopped. The class requires that a first-tier facility's acceptance of a unit cover the provenance of the parts, materials and firmware it contains and the authority for any change to the process that made it, and that the facility be able to state, for every unit already delivered, what it contains and how it was made, so that if a part, a lot or a process is later found compromised the affected units can be found. It requires that acceptance not rely only on a test the unit passes, because a deliberate alteration can be designed to pass the test. The provenance of components is the defense supply-chain class's to state at their source; the facility's requirement is that it can state what it fitted.

4.6 Allocation When Demand Exceeds Supply

When the war rate exceeds what the facilities can make, the output must be allocated: among the services, among theaters, between the force and the stockpile, and between the nation's own force and its allies. Each allocation decides which part of the force goes without. The class requires that the basis for allocating a first-tier capability be stated before it is needed, by the coordination class with the force, and that an allocation be made on that basis by a qualified human, on data whose integrity can be shown (Section 7.3). It does not state the allocation, which is a matter of national policy; it requires that the policy exist and that the data it acts on be trustworthy.

5. Design Basis Events: The Envelope

The events below are the class's design-basis envelope, each stated with the event postulated, the consequence threshold, the function the class requires and the duration for which it is required. Each is marked by the chain or chains it opens. Sections 5.1 and 5.4 are non-adversarial; Sections 5.3, 5.6 and 5.8 may be either, a counterfeit part as often coming from fraud for profit as from an adversary; Sections 5.2, 5.5 and 5.7 are adversarial. Inclusion is by national-security consequence severity on either axis, not likelihood (framework §4.1).

5.1 Loss of a Facility by Explosion, Fire or Natural Hazard

Postulated event an accidental explosion or fire in energetic materials, or a fire, flood, wind or earthquake, destroys or disables a first- or second-tier facility's production, loading or sustainment capability, or the units in it.

Consequence threshold loss of the facility's share of the qualified source count until it is restored and requalified, extending the time the force waits for the capability (Chain A); and, at an energetics facility, loss of life on site.

Required function the facility's protective functions against its own hazard act independently of any network an adversary could reach (Section 7.2); the facility states its production recovery time of Section 3.3; and the force's requirement can be met from other qualified sources or stock for that time.

Design-basis duration until the facility's rate is restored and its output accepted, or until the requirement is met by others.

The case is the fire at the Diehl group's site in the Lichterfelde district of Berlin on 3 May 2024, conferred by the framework as DBA-DIB-RE-1 and read under this category on the author's ruling of 14 October 2026 to follow the criminal record, held as the RECORD PRIMACY convention of the DBA Reference Event Register. The site belongs to the Diehl group and is operated by its subsidiary Diehl Metal Applications, which coats and stamps metal parts; the company said it does not belong to or supply Diehl Defence, which had had an office there until February 2024, and Diehl Defence makes the IRIS-T air-defense system supplied to Ukraine. The Berlin public prosecutor closed the proceedings in December 2024, finding no indication of intentional or negligent arson and pointing to an electrotechnical cause; the company's insurers had attributed the fire to technical problems in June 2024. The suspicion of sabotage reported from unnamed security sources by the Wall Street Journal in June 2024, among them that the fire began in an area few could enter and that video records of it were destroyed, and by The Telegraph is recorded as reporting, attributed to its sources; it does not decide the event's category, its actor or its tier. The instrument reads the event for the investigation it received and for what a defense group's other sites carry (*The Fire at the Diehl Plant in Berlin* F-1, F-2).

5.2 Sabotage of a Production Facility

Postulated event an adversary, directly or through people acting on its behalf, sets a fire, places a device or otherwise damages a facility, in peace or in a period short of war, in a way designed to be taken for an accident or to deny attribution.

Consequence threshold the loss of Section 5.1 (Chain A), together with the uncertainty that an unattributed event leaves about whether other facilities are being attacked, which delays the response that an attack would call for.

Required function the facility controls access to the areas whose loss would stop production and records who entered them; its records of those areas survive the event; an event at a first-tier facility is investigated from the start as possibly deliberate, with the agencies that would investigate an attack; and the event is reported so that the pattern across facilities can be seen. Beside the criminal investigation the facility and the coordination class hold a security-assessment track, which informs whether the event is to be treated as a possible attack while the criminal record is open and sets the protective posture of other facilities on it; the protective

response rests on that assessment and not only on the criminal finding (*The Fire at the Diehl Plant in Berlin* F-1, R-1; noted to the coordination class).

Design-basis duration the restoration of the facility, and the investigation until it can state whether the event was deliberate.

The class proposes as the case for this category, on the author's decision of 14 October 2026 and for the framework to confer, the fire and explosions that destroyed the Canadian Car and Foundry Company's plant at Kingsland, New Jersey, on 11 January 1917. The plant assembled artillery shells for Russia under contract, while the United States was still neutral. The fire set off some half a million shells over about four hours and destroyed the plant, and no one was killed, as reported. Its cause was disputed from the first. The claims against Germany for it, with those for the explosion at Black Tom in New York harbor in July 1916, came before the Mixed Claims Commission, United States and Germany, which decided both cases in Germany's favor on 16 October 1930, finding that the Kingsland fire was not caused by any German agent, and denied rehearing in 1931 and 1932. On the American agent's petition that the Commission had been misled by fraudulent, incomplete, collusive and false evidence, the Umpire ruled in December 1933 that, while the Commission could not reopen the cases for newly discovered evidence alone, it could reopen them to consider the charge that it had been misled by fraud and collusion. On 15 June 1939, the German Commissioner having withdrawn on 1 March 1939, the Umpire held that the United States had proved that fraud in the evidence presented by Germany had misled the Commission; the 1930 decision was set aside, Germany's liability was found established, and awards were made on 30 October 1939, as the Supreme Court of the United States recounts (*Z. & F. Assets Realization Corp. v. Hull*, 1941). The design-basis reading is that sabotage of a production plant passed as something other than enemy action for twenty-two years, nine of them under an adjudicated finding that no agent of the adversary's state had caused it, and was established only on a showing that the evidence that state had put before the tribunal was false. It is the category's case because the category postulates exactly that: deliberate damage designed to be taken for an accident or to deny attribution, and an uncertainty about whether the event was an attack that delays the response an attack would call for. The record is a tribunal's rather than a criminal court's, and it parts within itself; how the rule to follow the criminal record applies to it is for its instrument (Section 10.6). The worker named in contemporary reporting is not named here. The Diehl fire, which the class proposed for this category at v0.1, is read under Section 5.1 on the criminal record.

5.3 Loss of a Critical Input or Sole-Source Supplier

Postulated event the facility's supply of an input without which it cannot produce, an energetic material, a casting or forging, a component or a long-lead item of production equipment, is lost or delayed, through the loss of the supplier, an export restriction, a disruption of transport or a decision taken by a supplier under an adversary's law.

Consequence threshold the facility cannot produce at its rate, or at the war rate (Chain A); where the input has a single domestic source, every facility that depends on it is reached at once.

Required function the facility states its single points in the inputs and the stock or second source that covers each at the war rate (Section 4.3); and the class that governs the input states its resilience.

Design-basis duration until supply is restored or an alternative source is qualified.

5.4 Demand Beyond Capacity

Postulated event the force's consumption, its own or that of an ally it supplies, rises to a war rate beyond the facilities' capacity, so that the gap between the rate required and the rate produced persists.

Consequence threshold the force receives less than its design basis counts on for as long as the gap persists (Chain A), with no facility having been damaged; stockpiles are drawn down and are not restored.

Required function the surge rate and the time to reach it are stated for each first-tier capability (Section 4.2); new capacity is credited only from the date its output is accepted; and the allocation basis of Section 4.6 exists before the gap opens.

Design-basis duration from the opening of the gap until production meets the requirement or the requirement falls.

The record of 155 mm artillery ammunition in the United States since 2022 is the case, and it is noted rather than proposed. Production of about 14,000 rounds a month before the full-scale invasion of Ukraine was to rise to 100,000 a month by October 2025. In July 2026 the Department's Inspector General reported that the rate stood at about 36,000 rounds a month as of March 2026, after $4.5 billion spent from February 2022 to September 2025, and that a $469 million contractor-operated plant in Mesquite, Texas, built to run three lines making 30,000 projectile metal parts a month, had not produced a single shell meeting the Army's specifications, the plant having adapted older equipment to newer designs in what the Inspector General called an unproven process, as Stars and Stripes reported on 14 July 2026; the Inspector General put the risk of the shortfall as one to the United States, its allies and its partner nations. The design-basis reading is that the surge rate is set by the slowest input to qualify, and that capacity credited when it is funded or built, rather than when its output is accepted, overstates the rate the force will receive by years.

5.5 Exfiltration of Technical Data

Postulated event an adversary reaches a holding of a program's technical data package, at the facility or at a supplier or partner connected to it, through an intrusion into its networks, a person with access, or both, and copies the data over a period in which the copying is not detected.

Consequence threshold the loss of the design advantage of every unit of the program fielded and to be fielded, to the extent of the compromise scope of the holding reached (Chain X); the consequence is complete at the moment of the copy and is not reversed by anything done afterwards.

Required function the data holding and compromise scope of each first-tier program are stated, and the package is divided so that no single holding reaches more than its work needs (Section 4.4); copying is detectable within a stated time; and after an intrusion the facility can state what was reached, so that the program can assume the compromise of that scope and no more.

Design-basis duration the service life of the program and of any successor whose design inherits the data.

The case is the theft of data on the C-17 transport aircraft and on fighter aircraft from defense contractors between 2008 and 2014, conferred by the framework as DBA-DIB-RE-3 (framework §9.1). On the author's ruling of 14 October 2026 that a state's covert theft operations are military actions, its instrument records it as military action without the military-action tier, denial not being determined (*Design Basis Layers* §6.4). On the Department of Justice's account, a conspiracy that ran from October 2008 to March 2014 reached computers belonging to defense contractors, including computers belonging to Boeing in Orange County, California, and took sensitive military and export-controlled data on the C-17 and on certain fighter aircraft produced for the United States military. Two co-conspirators in China, whom the Department described as hackers of the People's Liberation Army Air Force, carried out the intrusions; a businessman in Canada directed them to the people, companies and files to target, using lists of the directories they had reached, and translated some of what was taken; and the data was sent to China. The businessman, who on the plea record acted for financial gain and helped write reports on the data's value for those to whom it was sold, was arrested in Canada in July 2014, waived extradition in February 2016, pleaded guilty in March 2016 to conspiring to gain unauthorized access to a protected computer and to violate the Arms Export Control Act, and was sentenced in July 2016 in the Central District of California to forty-six months in prison. The complaint, as reported at the time, quoted the conspirators' own report as claiming about 630,000 files, some 65 gigabytes, on the C-17 alone; that figure is their claim, not a finding. The design-basis reading is that the data of a program in production was taken over more than five years, guided by someone who knew the industry and which files mattered, and that the consequence, whatever the extent finally established, could not be recovered once the copy was made. The instrument states the government's account as the government's and makes no finding about what was taken beyond what its sources establish.

5.6 Corruption of the Output

Postulated event a unit leaves the facility other than what it is said to be, through a counterfeit or altered part accepted from the supply chain, firmware that is not the intended firmware, or a

change to the production process made without authority, so that it fails or can be made to fail in service.

Consequence threshold the failure of fielded units when they are needed, in a number set by how many units the corruption reached before it was found (Chain I), and the loss of confidence in every unit of the same lot, make or program until they are cleared.

Required function acceptance covers provenance and the authority for process changes, and does not rely only on a test the unit passes; and the facility can state what every delivered unit contains (Section 4.5).

Design-basis duration the service life of the units affected.

The firmware and electronics in a unit are products of the defense supply-chain class and of the information-technology series' supply-chain class (*Software, Hardware and Product Supply Chain*, DBA-IT-SW-FC3); this class requires that the facility be able to state what it fitted and on whose authority it changed how it built.

5.7 Cyber Attack on Production Systems

Postulated event an adversary reaches the control systems of a facility's production, loading or test equipment and causes it to fail or be damaged, prevents its safe shutdown, or changes a process so that the output is defective.

Consequence threshold physical damage to the production capability (Chain A), harm to people where the equipment is hazardous, and, where a process is changed, corrupted output (Chain I).

Required function the safe shutdown of hazardous equipment does not depend on systems reachable from the facility's business network (Section 7.2); changes to process settings are made only through a governed act and recorded; and the facility states its time to restore each item of equipment whose loss would stop production.

Design-basis duration until the equipment is restored and the output since the attack has been cleared.

The attack on a German steel mill that *Long-Lead Critical Components* reads at its §5.5 is the precedent for physical damage by this route, and this class cites it there rather than restating it.

5.8 Loss of the Cleared and Skilled Workforce

Postulated event the facility cannot staff its production, loading, sustainment or acceptance functions, through the loss of people with the skills or the clearances the work requires, the time to clear or train replacements, illness, evacuation, or the recruitment of its people by an adversary.

Consequence threshold a fall in the facility's rate (Chain A); and, where a person with access is recruited or coerced, a path for Chain X or Chain I.

Required function the facility states the staffing on which its rate and surge depend and the time to clear and train replacements; and the access of each person to first-tier data and processes is limited to what that person's work needs. Under a positive applicability determination it states also the workforce, its housing and its transport within a campaign's reach, with their shelter and the effect of their loss on the rate (*Ukrainian Strikes on Russian Defense Production* F-2, R-2).

Design-basis duration the duration of the shortage, and for access, the period of access.

Table 5-1. Facility-class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Framework source
Loss of a facility by explosion, fire or natural hazard	Production, loading or sustainment capability destroyed	Chain A	§4.1 candidate (loss of a sole-source capability)
Sabotage of a production facility	Deliberate damage designed to pass as an accident	Chain A	§4.1 candidate; §7; developed here
Loss of a critical input or sole-source supplier	Energetics, castings, components or production equipment lost	Chain A	§2.1, §3.3; DBA-CM
Demand beyond capacity	War rate exceeds what the facilities can make	Chain A	§2.1, §3.5; developed here
Exfiltration of technical data	A holding of the technical data package copied	Chain X	§1.5, §3.6; §4.1 candidate (from RD)
Corruption of the output	Counterfeit or altered part, firmware or process	Chain I	§3.2, §5; §4.1 candidate (from SC)
Cyber attack on production systems	Equipment damaged or process changed	Chain A, Chain I	§5; developed here
Loss of the cleared and skilled workforce	Production or acceptance unstaffed, or access abused	Chain A, Chain X, Chain I	developed here

6. Military-Action Extension: DBA-MA for Defense Production

6.1 MA Extension Applicability Determination

Every first- and second-tier facility in the class, on either axis, performs the facility-level applicability determination the series makes mandatory, under the two-factor test of geolocation

and strategic value, with strategic value driven by irreplaceability and by the sensitivity of what the facility holds (framework §7). The framework calls this the sector where the military-action extension is most native, because it is the base a military action is intended to defeat or deny, and states a strong sector-level presumption toward applicability for sole-source production (framework §7). For this class geolocation includes the reach of the weapons an adversary can bring to bear, which the record of Section 6.3 shows extends far from any front, and the reach of its intelligence services into the facility's networks and workforce, which has no distance at all. The determination is recorded, it is time-varying and re-triggered on the indicators the DBA-MA parent framework defines, and any scope reduction carries a recorded basis (Section 6.9). The determination is made for every site that carries a defense group's name, whatever the site makes, so that a site with a known defense owner states its exposure to sabotage (*The Fire at the Diehl Plant in Berlin* F-2, R-2).

6.2 The Four Characteristics Applied to the Class

Applied to defense production, the four military-action characteristics take specific form, in the canonical statement of FD-DBL (*Design Basis Layers*), at its §4.4. Intentionality: the adversary chooses the facilities whose output its own forces most need to see reduced, and the data whose possession most improves its own weapons or defeats the defender's. Persistence: the adversary strikes production repeatedly over a campaign, and takes data over years, as the record of Section 5.5 shows. Precision: the adversary chooses the sole-source facility, the single input, the holding with the widest scope and the moment when stocks are drawn down. Denial: the adversary destroys or disables the capability so that it cannot be restored within the time the conflict lasts, or, on the confidentiality axis, takes from the defender the advantage the program was built to give. The framework names persistence and denial as the dominant characteristics for the sector, with pre-positioning in the supply chain and the compromise of a capability before it is needed (framework §7).

Under the author's rulings of 10 and 13 October 2026, the military-action tier is not met without denial, and all four characteristics must be established on the record; a characteristic not determined counts as not present (*Design Basis Layers* §6.4). On the author's ruling of 14 October 2026 a state's surveillance and covert theft operations are military actions, so DBA-DIB-RE-3 records the C-17 theft as military action without the tier; DBA-DIB-RE-2 records the strikes on Alabuga the same way, denial not being determined; and DBA-DIB-RE-1 does not meet the threshold, no adversary being established on the criminal record.

6.3 Production Struck at Depth, as the Governing MA Scenario

The governing military-action scenario is a campaign against the defender's defense production during a war, by long-range strikes on its plants, its energetics and its depots, wherever they are within reach, combined with sabotage of plants beyond reach. It opens Chain A at the war rate, when demand is highest and stocks are lowest. No credit is taken for distance from the front, for a facility's obscurity, or for stocks and sources the same campaign can reach.

The extension requires three things. It requires that the war rate of each first-tier capability be met from facilities and inputs of which no single loss, and no loss the campaign can impose within the time stocks last, reduces production below the force's requirement. It requires that the protective functions of energetics facilities limit the loss from a strike to the building struck, by the separation and protection those facilities already carry against their own hazard (Section 7.2). And it requires that the stocks the force draws on while production is restored be held where the campaign cannot reach them all. Protection is designed against the campaign's weapon: physical protection over the buildings whose loss stops production is designed against the weapons the campaign is expected to use and credited only against them, because at Alabuga the production halls fitted in 2023 with anti-drone cages were not damaged in the imagery of the 2024 strike while a dormitory nearby was (*Ukrainian Strikes on Russian Defense Production* F-1, R-1). And the workforce, its housing and its transport are within the scenario (Section 5.8; *Ukrainian Strikes on Russian Defense Production* F-2).

The empirical case is the campaign of Ukrainian strikes on Russian defense production, conferred by the framework as DBA-DIB-RE-2 (framework §9.1), read from the side of the plant struck, and bounded by its instrument to the strikes on one site on 2 April 2024 and 23 April 2025. On 23 April 2025 the General Staff of Ukraine's armed forces said that its Unmanned Systems Forces, with other branches, had struck a plant in the Alabuga special economic zone in the Republic of Tatarstan, more than 1,000 kilometers from Ukraine's border, which it said produced up to 300 Shahed-type and Geran drones a day, reporting hits and explosions in the target area and that the extent of the damage was being assessed; the Russian Ministry of Defense said that air defenses had shot down one drone over Tatarstan and did not mention the plant, as the Kyiv Independent reported. A strike on the same site on 2 April 2024 was assessed from commercial satellite imagery by the Institute for Science and International Security, which found damage to dormitories and none visible to the production halls, and Ukraine has claimed strikes on other Russian defense plants and depots. The design-basis reading is that a belligerent judged a production facility far from the front to be within reach and worth striking repeatedly, and that the defender of such a facility must plan for it to be struck. That is the scenario the extension postulates for the nation's own facilities. The record is a belligerent's claims and its adversary's statements, and the extent of the damage is not established by either; the instrument states each party's account as its own, bounds the campaign it reads, and makes no finding of effect beyond what its sources establish.

6.4 Compromise Before the Conflict

The second scenario is on the confidentiality-and-integrity axis, and it is the one the framework calls the sector's own: the deliberate compromise of a capability before it is needed (framework §7). An adversary that has taken a program's data in peace enters a conflict knowing the program's weaknesses; an adversary that has placed an altered part or firmware in the program's units can choose when they fail. Neither requires a strike, and neither is visible until it is used. The class requires a first-tier program's design basis to assume the compromise of any holding

that has been reached, to the extent of its compromise scope, and to state how the program would fight with that assumption; and it requires that the provenance coverage of Section 3.3 be enough to find the units an implant reached once it is discovered.

6.5 The Compound Scenario

The framework names as a candidate bounding event the compound kinetic-plus-cyber attack that denies a capability while compromising its replacement (framework §4.1). For this class it is a strike on a facility, timed to a campaign, while the adversary holds access to the networks of the facilities that would replace it, so that the replacement's process can be changed or its output corrupted. Against it the class relies on the separation of process changes from networks an adversary can reach (Section 5.7), on acceptance that covers provenance and process authority (Section 4.5), and on qualified sources whose networks are not reached by the same intrusion.

6.6 Cross-Sector Interaction

Under military action the class's outward consequence is the capability of the force itself (framework §6.2), and, on the confidentiality axis, a gain to the adversary that no other sector produces. Coming in, the class depends on Critical Manufacturing for its long-lead production equipment and materials, on energy for its processes and on the electric distribution facilities the framework names through the defense-critical-load finding, on transportation for moving its inputs and its output to the force, on the chemical sector for the precursors of energetics, and on information technology for the networks that hold its data (framework §6.1). An adversary need not strike a defense plant to stop it: a campaign against its energy, its precursors or the port through which its output moves can be enough (Section 8.2).

6.7 The Adequacy Result: Two Conditions

Because the framework refuses to collapse the two axes into one (framework §3.6), the test of adequacy is two tests, and a facility meets the class design basis only if it meets both.

The first is what this document calls the Sustained Production Condition. When demand is at the war rate the design basis postulates, and any one facility, any one critical input source and any one route are lost, the force receives each first-tier capability at the rate its own design basis requires, within the time its stocks cover, from qualified facilities and inputs the adversary has not reached, by routes it has not cut. The condition bounds Chain A by the surge rate, the qualified source count and the stocks, and it is met only by capacity whose output has been accepted.

The second is what this document calls the Bounded Compromise Condition. For each first-tier program, the copying of any one holding reaches no more than its stated compromise scope; the copying is detectable within the stated time; and the facility can state, for every delivered unit, what it contains, so that if any part, lot or process is found compromised the units it reached can be found and the corruption bounded. The condition does not bound the consequence of a

compromise once it has occurred, which is not recoverable; it bounds its scope. It bounds Chain X by the division of the holdings and Chain I by the provenance coverage.

The protective functions on which an energetics facility relies are those that keep its own product from destroying it: deluge and suppression systems, the interlocks of remote operations, and the limits on quantities and on the processes that heat, press or mix energetic materials. These are to be locally sovereign as FD-LS defines the property where they are protective functions in that sense (*Local Sovereignty* §3.2): a measurement against a setpoint, colocated with the hazard, which no system an attacker could reach can stop, delay or trigger, and whose settings change only through a governed act. Local Sovereignty governs and is not restated here. It describes what protects the facility when its control systems have failed; it does not decide whether any action needs human authorization (*Local Sovereignty* §4). It has nothing to say about the confidentiality axis, which no physical protection can bound.

6.8 Adequacy and the Declared Residual

In the non-adversarial domain, the Sustained Production Condition is achievable with the Section 4 requirements: a stated surge rate, stated single points with sized stocks or second sources, and capacity credited only when its output is accepted. Against a deliberate adversary, adequacy has limits on both axes.

Three residuals are declared, in the output form DBA-MA-DiD (*Defense in Depth*) sets at its §3.5. First, a campaign that destroys production faster than the reachable facilities and stocks can replace leaves the force short; beyond the design campaign the residual is a shortfall measured in rate and time, and the response is allied production, the allocation of Section 4.6 and the dispersal of new capacity. Second, data already copied is a residual with no end in time; its measure is the compromise scope already reached, and the response is to assume it, to change the design of the program’s successors where the compromise reached, and to bound future holdings. Third, an implant that passes acceptance is a residual in fielded units; its measure is the units it reached, and the response is the provenance coverage that allows them to be found. These residuals bound the class adequacy claim.

6.9 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter, such as a surge rate credited from capacity whose output has not been accepted, a qualified source counted that is under an adversary’s control, a stock credited that the campaign can reach, a facility credited as safe by its distance from any front, a holding credited as protected because it is certified rather than because its compromise scope has been bounded, or an acceptance test credited as detecting a deliberate alteration it was not designed to detect, must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework does not yet state this principle for the sector, and Sections 9.1 and 10.9 record it for the framework to adopt. Physical protection

credited against a weapon it was not designed against is such a relaxation (*Ukrainian Strikes on Russian Defense Production* R-1).

7. The Dual Design Basis, Stated in Full

7.1 The Two Design Bases

The facility carries two design bases at once: a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). They interact at the point developed in Section 7.4. The two design bases are not the two consequence axes: both design bases apply on both axes.

7.2 The Physical Design Basis: The Determinism Posture

The physical design basis is the set of protective functions the facility relies on to keep its people and its capability safe independently of its networked systems, and the controls on the quality of what it builds: the explosives-safety separations, protections and quantity limits of energetics facilities; the deluge, suppression and interlocks that act on their hazards; the safe shutdown of heavy production and test equipment; and the acceptance of each unit. A deluge system that acts on its own detection of a flash, with no networked controller able to inhibit or command it, is a candidate for a locally sovereign function. A function meets *Local Sovereignty* §3.2 only if no system an attacker could reach can stop, delay or trigger it, or change its settings except through a governed act.

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Controllers that behave predictably within a stated range of inputs, and that have been formally tested in the deployed configuration, are outside the Command Broker obligation for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). That is a statement of scope, not of placement; they are outside the obligation only because, and only while, their behavior is bounded by that showing (*The Bright Line* §4.5). The exemption rests on the configuration as installed and tested, and it lapses for a configuration changed outside a governed act (Determination of 8 October 2026, A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out). The framework adds the components-integrity precondition that binds here because supply-chain compromise is the sector’s governing threat: a compromised or unverified AI or firmware component does not meet the carve-out, whatever its apparent behavior, and each of its actions is judged on its own consequence (framework §5).

7.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems are entering the class: generative design of parts and tooling; autonomous control of machining, loading and assembly; machine inspection in acceptance; predictive maintenance that can lock out equipment; production scheduling and the

allocation of output; and assistants with access to the technical data package. The framework names these and places the Bright Line for any action of a component within the scope of *Layer Decomposition and Determinism* §4.3 whose incorrect or compromised performance could corrupt a design, a manufacturing process or an acquisition decision at or above the severity threshold (framework §5). These systems are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*); CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. The coordination class, CorpCo, is the locus at which the sector's placement and obligation are implemented (framework §3.5). Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3).

Four cases illustrate the application, and none of them is a placement by component. A system within scope that releases technical data, by sending a file to a supplier, answering a query from the data or passing the data to a model outside the facility's control, takes an action with each release, placed on the consequence of the release, which on the confidentiality axis is the compromise scope of what is released and cannot be undone; a release whose consequence can reach the threshold requires the authorization of the qualified human the coordination class names as Command Broker, given before the release because no review after it can recover the data. A release within a set of recipients and data already authorized may be discharged in advance under FD-BL-D1 (*Mode of Discharge*); a change to the set itself, being a change to the pre-authorized envelope, is excluded from advance discharge by that determination's rule R3. A system within scope that accepts a unit on the basis of machine inspection takes, with the acceptance, one action placed on the consequence of the unit's failure in service. A system within scope that changes a process setting takes an action placed on the consequence of the change, which can be the corruption of every unit made under it. And a system within scope that allocates first-tier output takes an action placed on the consequence to the part of the force that goes without (Section 4.6).

Any output on which such an action is grounded carries the indication-integrity obligation of FD-BL-D2 (*Indication Integrity*): an inspection result, an allocation computed from requirement and inventory data, or a provenance record that may have been altered is not a basis for authorization until it can be corroborated. Where it cannot be corroborated through an independent path, the action it bears on does not proceed without in-the-moment Command Broker authorization, and no placement changes (framework §5). In this class the adversary that can alter the basis is the same adversary that seeks to corrupt the output, which is why the obligation binds here.

7.4 The Interaction Point: Release and Acceptance Under a Campaign

The two design bases meet where the facility's systems release data or accept output while an adversary is inside its networks. The facility's systems must be architecturally incapable of executing an action requiring authorization that originates from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization. That

obligation reaches each such action whose consequence meets the threshold, and the actions most likely to do so are the release of first-tier technical data, the acceptance of a first-tier unit, a change to the process that makes one, and the allocation of first-tier output; each is placed on its own consequence. The physical design basis keeps the facility's people and capability safe; the division of the holdings and the provenance coverage bound the compromise; the AI-governance design basis guarantees that no proposing system can release what cannot be recovered, or accept what may not be what it seems, where the consequence requires it, without a qualified human authorization the architecture can enforce.

7.5 Scope Boundary

This instrument states the physical design basis in full and names the AI-governance design basis without restating it. The Foundational Definitions own the Bright Line, its scope and its obligation; the Command Broker family owns the Command Broker; the coordination class owns the implementation of placement in the sector, mobilization and allocation policy; the supply-chain and research classes own the protection of their own holdings and components; and Critical Manufacturing owns the long-lead production equipment and materials. This instrument applies them to the facilities that produce and sustain defense materiel.

8. Cross-Sector Dependencies and Regulatory Context

8.1 What Depends on the Class: The Capability of the Force

The class's outward consequence is the nation's ability to equip and sustain its force and the forces of the allies it supplies (framework §6.2). The record since 2022 shows that the allies' consumption is part of the demand the class must meet (Section 5.4). On the confidentiality axis the outward consequence is the adversary's gain, which reaches every unit of a compromised program wherever it is fielded, including units sold to allies.

8.2 What the Facility Depends On

The facility depends on Critical Manufacturing for long-lead production equipment, heavy castings and forgings and specialty materials, and through it on the RT_HVLLC that bounds their replacement (*Long-Lead Critical Components* §3.3); on the defense supply-chain class for components and microelectronics; on the chemical series for the precursors of propellants and explosives, among them nitric acid, whose production the chemical series governs (*Chemical Production and Synthesis*, DBA-CHEM-PR-FC1); on energy for its processes, through electric distribution facilities the framework elevates by the defense-critical-load finding (framework §6.1); on transportation for its inputs and for the movement of its output to the force through strategic seaports and corridors, which the transportation series governs (*Nodes and Terminals*, DBA-TRAN-ND-FC1); and on information technology for the networks that hold its data (framework §6.1).

8.3 The Boundaries with the Partner Classes and Series

Five boundaries are drawn. With *Defense Supply Chain* (DBA-DIB-SC-FC2): components, materials and microelectronics, and the protection of the data holdings at the suppliers that make them, are that class's; the facility's statement of its holdings at its suppliers, and its acceptance of what they supply, are this class's. With *Research, Development, Test and Evaluation* (DBA-DIB-RD-FC3): the data a facility holds in order to build is this class's and the data held in order to design and test is that class's; where a prime holds both at one site, both classes apply, and the exfiltration of Section 5.5 may be read in either. With the coordination class, CorpCo: mobilization, surge planning, allocation policy and the implementation of Bright Line placement in the sector are its. With Critical Manufacturing: the boundary runs where a component enters a defense system, the general long-lead and concentration analysis being that series' and the defense-specific integrity and classification analysis this one's (framework §6.3). With the surveillance sibling standard: covert-influence operations against the industrial base are that standard's, and exfiltration and corruption are this sector's (framework §6.3).

8.4 Regulatory and Institutional Context

The class is governed less by safety regulation than by security and acquisition regulation (framework §2.1). The Department of Defense is the Sector Risk Management Agency, the role the framework calls Sector-Specific Agency. Contractors that hold covered defense information are bound by the Defense Federal Acquisition Regulation Supplement clause on safeguarding that information and reporting cyber incidents (DFARS 252.204-7012), which requires the protections of NIST SP 800-171. The Cybersecurity Maturity Model Certification program, whose program rule (32 CFR Part 170) took effect on 16 December 2024 and whose acquisition rule took effect on 10 November 2025, with phased implementation, verifies those protections. The export of defense articles and of their technical data is controlled by the International Traffic in Arms Regulations (22 CFR Parts 120 to 130), and access to classified information by the National Industrial Security Program (32 CFR Part 117). The Defense Production Act gives the government the authority to require priority performance of contracts and orders under the Defense Priorities and Allocations System regulation (15 CFR Part 700) and, under Title III, to finance the expansion of domestic capacity. Explosives safety at energetics facilities is governed by the Department's explosives-safety standards under the Department of Defense Explosives Safety Board. Foreign investment in the class's facilities is screened by the Committee on Foreign Investment in the United States. The National Defense Industrial Strategy of January 2024 states the Department's policy for the base's resilience.

The regulatory gap is the framework's (framework §2.3). The apparatus measures security in controls and certifications. No instrument states the war rate the nation must be shown able to reach and the time to reach it, crediting only accepted capacity; no instrument treats a program's data holdings and their compromise scope as a design-basis quantity; and no instrument requires that the units a compromise reached be findable. This class states those as design-basis requirements for the tiers that carry the load.

9. Inheritance Map and Anticipated Derivatives

9.1 The Inheritance Map

The series requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines locally. Facility-class instruments add the defense-function-specific consequence chain on both the availability and the confidentiality-and-integrity axes (framework §8.1). Table 9-1 is the map for this class. Where the framework is a skeleton, the element is developed here and recorded for the framework to take up.

Table 9-1. Inheritance map for DBA-DIB-DP-FC1.

Element	Source in the framework	Inheritance mode
Function-primary placement, DP function class	§1.3, §3.1, §3.2	Inherited unchanged
Classification and sole-source sub-index	§3.1, §8.2	Inherited with adaptation; three tiers stated on each axis separately (Section 2.2)
Design-basis principles (consequence severity on either axis, not likelihood)	§4.1	Inherited unchanged
Dual-axis consequence	§1.5, §3.6	Inherited; the confidentiality-and-integrity axis separated into Chain X and Chain I (Section 3.2)
RT_HVLLC on the availability axis	§3.6; DBA-CM-LL-FC1	Inherited; applied as the production recovery time (Section 3.3)
Non-recoverable consequence measure	§3.6 (to be developed, §9)	Locally defined as compromise scope (Section 3.3); for the framework to take up
Event categories	§4.1 (candidates)	Developed here ahead of the framework; for the framework to take up
Bright Line and Command Broker architecture; components-integrity precondition	§5; FD-BL; Command Broker family	Inherited with adaptation: release of technical data placed on its compromise scope (Section 7.3)
The production line and the technical data package	§3.2	Locally defined (Section 4)

Element	Source in the framework	Inheritance mode
Sustained Production and Bounded Compromise Conditions	§7 (skeleton)	Locally defined (Section 6.7)
Declared residuals and conditionality check	not yet in the framework	Locally defined; recorded for the framework (Sections 6.8, 6.9)
Boundaries with Critical Manufacturing and the surveillance standard	§6.3	Inherited unchanged and applied (Section 8.3)

9.2 Anticipated Derivatives

Facility-, program- and yard-specific design bases inherit this class document by reference and add only what is locally specific, each carrying a recorded applicability determination for any scope it reduces. The three reference events this class proposed at v0.1, on the author's decision of 14 October 2026, are conferred by the framework at §9.1: DBA-DIB-RE-1, *The Fire at the Diehl Plant in Berlin*, 3 May 2024, read under Section 5.1 on the author's ruling of 14 October 2026 to follow the criminal record, held as the RECORD PRIMACY convention of the DBA Reference Event Register; DBA-DIB-RE-2, *Ukrainian Strikes on Russian Defense Production*, 2024 to 2025 (Section 6.3); and DBA-DIB-RE-3, *The Theft of C-17 and Fighter Aircraft Data From Defense Contractors*, 2008 to 2014 (Section 5.5). Their instruments were written by Project 11 on 14 October 2026, and this revision takes in their erratum and six recommendations. For Section 5.2, left without a case when the Diehl fire moved to Section 5.1, this revision proposes a fourth event, on the author's decision of 14 October 2026, for the framework to confer: the fire and explosions at the Canadian Car and Foundry Company's shell plant at Kingsland, New Jersey, on 11 January 1917 (Section 5.2). The record of 155 mm production since 2022 is noted and not proposed, being a condition still running whose account is the Inspector General's (Section 5.4). The facility-class use case for the governing military-action scenario of Section 6.3 is DBA-DIB-UC-1, issued under this revision in the fifth wave of the plan.

10. Gaps and Open Items

10.1 Quantitative Design Parameters

The six parameters of Section 3.3 are stated without values. Their values for a facility or program are to be validated against the Department's industrial-base assessments, the services' munitions requirements, the contractors' records of their data holdings and the assessments made under the requirements of Section 8.4. Many of those values are themselves controlled or classified, and a facility's design basis that states them must be held at the level of what it states. This instrument is written to be published, and states none.

10.2 The Non-Recoverable Consequence Measure

The framework introduces a non-recoverable confidentiality-and-integrity consequence category whose measure is the scope of capability compromised rather than a duration, and records its development as an open item (framework §3.6, §9). This instrument states it for the production function as the compromise scope of a data holding and the provenance coverage of delivered units (Section 3.3), and separates the axis into Chain X and Chain I (Section 3.2). Whether the framework adopts the two parameters, and the separation, is for the framework. The supply-chain and research classes will state the same measure for their own functions, and the framework should reconcile the three.

10.3 Two Axes in One Facility

The class states a facility's tier on each axis separately and requires it to meet an adequacy condition on each (Sections 2.2, 6.7). A facility can be third-tier on one axis and first-tier on the other: a machine shop that makes a common part for many programs and holds a fragment of each program's data, or a sole-source energetics plant that holds little data of consequence. The framework's sub-index is stated as one; this class asks the framework to state it as two.

10.4 Local Sovereignty in the DBA-MA Sense

Section 6.7 relies on locally sovereign protective functions at energetics facilities, as FD-LS defines the property. Whether the DBA-MA parent framework's local-sovereignty principle is the same thing FD-LS defines remains the open question the other classes record.

10.5 The Agency's Name

The framework names the Department of Defense the Sector-Specific Agency (framework §1.4, §2.1). The role has since been renamed Sector Risk Management Agency, and this instrument uses the current name. The framework's usage is the framework's to conform at its next revision.

10.6 The Reference Events

The three reference events this class proposed are conferred by the framework at §9.1 as DBA-DIB-RE-1 to DBA-DIB-RE-3, and their instruments were written by Project 11 on 14 October 2026. DBA-DIB-RE-1, at v0.2, reads the Diehl fire under Section 5.1 on the author's ruling of 14 October 2026 to follow the criminal record, held as the RECORD PRIMACY convention of the DBA Reference Event Register, the criminal record closed in December 2024 with no arson found, and does not meet the threshold; DBA-DIB-RE-2 bounds the campaign to the two strikes on Alabuga and records them as military action without the tier, denial not determined, the record open; and DBA-DIB-RE-3 records the C-17 theft as military action without the tier, on the Department of Justice's account and the court record. The instruments name no individual. For Section 5.2, on the author's decision of 14 October 2026 to seek a new event, this class proposes the Kingsland fire of 11 January 1917, for the framework to confer and Project 11 to write. Three points are for the instrument. The record is the Mixed Claims Commission's, an

international tribunal, not a criminal court, and its own findings part: dismissal in 1930, liability in 1939 on a finding that the evidence presented by Germany was fraudulent; whether the rule to follow the criminal record reaches a tribunal's award, and which of the tribunal's findings it follows, is for Project 11 and, as the rule's home, Project 0. On the Commission's finding of 1939 the act engaged a state's liability, in a country then neutral, and its category and tier are the instrument's to determine under the author's rulings. And the instrument should rest on the Commission's opinions and awards and on the Supreme Court's account of them, state each as its own, and name no individual. This revision takes in the erratum at Sections 1.3, 5.1 and 5.2 and the recommendations at Sections 3.3, 4.4, 5.2, 5.8, 6.1, 6.3 and 6.9. The use case for the governing scenario of Section 6.3 is DBA-DIB-UC-1.

10.7 The Register Row

The DBA Facility Class Register now records this class at v0.1 as an issued working draft under the framework at v0.3; Project 0 entered the row on 14 October 2026.

10.8 The Framework's Date

The framework's cover reads August 2026 at v0.3, while its revision history records v0.3 as made in October 2026. The cover date is the framework's to correct at its next revision, which the conferral will be.

10.9 What This Class Notes to Other Instruments

Several elements of this instrument belong to other instruments, and this instrument records them as noted without stating them for those instruments. To the framework: the separation of the confidentiality-and-integrity axis into Chain X and Chain I (Section 3.2), the six parameters and the compromise scope as its non-recoverable measure (Section 3.3, Section 10.2), the event categories of Section 5, the tiers stated on each axis (Sections 2.2, 10.3), the two adequacy conditions of Section 6.7, the declared residuals of Section 6.8, the conditionality principle of Section 6.9, the agency's name (Section 10.5), and the cover date (Section 10.8); the conferral section it also asked for is now framework §9.1, and the fourth event proposed for Section 5.2 is the framework's to confer. To *Long-Lead Critical Components*: that this class applies its RT_HVLLC parameters to defense production equipment as the production recovery time (Section 3.3). To *Defense Supply Chain* and *Research, Development, Test and Evaluation*, when written: the boundaries of Section 8.3 and the measure of Section 10.2, and to the former *The Theft of C-17 and Fighter Aircraft Data From Defense Contractors* R-2, that the people with access are part of the holding. To the coordination class, when written: *The Fire at the Diehl Plant in Berlin* R-1, the security-assessment track beside the criminal one. To the chemical and transportation series: the dependences of Section 8.2. The register row is entered (Section 10.7).

References

DBA-DIB Series and Parent Framework

Defense Industrial Base Sector Framework (DBA-DIB_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

DBA-MA Parent Framework. Eclectic Technologies. Source of the design-basis-under-military-action methodology. FD-DBL (*Design Basis Layers*), held for the edition following the October 2026 Edition, is the canonical statement of the four military-action characteristics (§4.4, §6.4).

Defense in Depth (DBA-MA-DiD). Eclectic Technologies, 2026. Declared-residual output form (§3.5).

Long-Lead Critical Components (DBA-CM-LL-FC1), *Chemical Production and Synthesis* (DBA-CHEM-PR-FC1), *Nodes and Terminals* (DBA-TRAN-ND-FC1), *Care Delivery* (DBA-HPH-CD-FC1) and *Software, Hardware and Product Supply Chain* (DBA-IT-SW-FC3). Eclectic Technologies, 2026. The availability axis and the steel-mill precedent, the precursors of energetics, the movement of output, the structural model, and the supply chain of fitted firmware.

The Fire at the Diehl Plant in Berlin (DBA-DIB-RE-1), *Ukrainian Strikes on Russian Defense Production* (DBA-DIB-RE-2) and *The Theft of C-17 and Fighter Aircraft Data From Defense Contractors* (DBA-DIB-RE-3). Eclectic Technologies, 2026. Held by Project 11; conferred at Defense Industrial Base Sector Framework §9.1 and cited by short title and finding or recommendation number. The RECORD PRIMACY convention of the DBA Reference Event Register, Project 0.

Facility-Class Types Are Physical, Coordination and Logical, author's determination of 30 September 2026, and *A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out*, author's determination of 8 October 2026, both recorded by Project 0. *Closing the Seven Sectors*, working plan of 13 October 2026, Project 9.

Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

Regulatory, Standards and Guidance

Defense Federal Acquisition Regulation Supplement, 252.204-7012, Safeguarding Covered Defense Information and Cyber Incident Reporting. National Institute of Standards and Technology. SP 800-171, Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations. Cybersecurity Maturity Model Certification Program, 32 CFR Part 170, and the acquisition rule implementing it in the Defense Federal Acquisition Regulation Supplement,

effective 10 November 2025. International Traffic in Arms Regulations, 22 CFR Parts 120 to 130. National Industrial Security Program Operating Manual, 32 CFR Part 117. Defense Production Act of 1950, as amended, Titles I and III; Defense Priorities and Allocations System, 15 CFR Part 700. Department of Defense Explosives Safety Board and the Department's explosives-safety standards. Department of Defense. *National Defense Industrial Strategy*, January 2024.

Events and Investigations

The Wall Street Journal, reporting of the investigation of the fire at the Diehl group's plant in Berlin, June 2024, and *The Telegraph*, later reporting of the same investigation. The company's statements on the cause of the fire, as reported.

General Staff of the Armed Forces of Ukraine, statement of 23 April 2025 on the strike on the Alabuga plant, and the Russian Ministry of Defense's statement of the same day, as reported by the *Kyiv Independent*, 23 April 2025; and reporting of the strike on the same site in April 2024.

United States Department of Justice. *Chinese National Who Conspired to Hack into U.S. Defense Contractors' Systems Sentenced to 46 Months in Federal Prison*, 13 July 2016, and the record of the case in the United States District Court for the Central District of California; contemporary reporting of the complaint.

Office of Inspector General of the Department. Report on the production of 155 mm artillery ammunition, July 2026, as reported by *Stars and Stripes*, 14 July 2026.

Mixed Claims Commission, United States and Germany. *Lehigh Valley Railroad Company, Agency of Canadian Car and Foundry Company, Limited, and Various Underwriters (United States) v. Germany* (Sabotage Cases), *Reports of International Arbitral Awards*, vol. VIII: the decision of 16 October 1930; the denials of rehearing of 1931 and 1932; the Umpire's decision of December 1933; the decision of 15 June 1939; and the awards of 30 October 1939. Supreme Court of the United States. *Z. & F. Assets Realization Corp. v. Hull*, 311 U.S. 470 (1941). Contemporary reporting of the Kingsland fire, January 1917.

Revision History

Version	Date	Author	Description
0.1 — DRAFT	October 2026	Tim Roxey	Initial issue. First facility-class instrument in the DBA-DIB series, typed physical under the author's determination of 30 September 2026, in the ten-section structure of DBA-HPH-CD-FC1 and DBA-CHEM-PR-FC1. Written under the plan Closing the Seven Sectors, wave 3, on the author's decisions of 14 October 2026: Defense Production and Sustainment as the sector's first class; three reference events proposed, the fire at the Diehl group's plant in Berlin (2024), the Ukrainian strikes on Russian defense production (from 2024) and the theft of C-17 and fighter data from

Version	Date	Author	Description
			defense contractors (2008 to 2014). States both of the framework's consequence axes for the production function, separates the confidentiality-and-integrity axis into an exfiltration chain and an integrity chain, applies RT_HVLLC from DBA-CM-LL-FC1 as the production recovery time, states compromise scope as the non-recoverable measure, and states two adequacy conditions, Sustained Production and Bounded Compromise. Checked before issue by an independent review agent, not a person, against primary sources and for internal consistency; its substantive findings were taken in, among them the volume of C-17 data stated as the conspirators' own claim quoted in the complaint, an unsupported statement attributed to the company removed, and the later reporting on the Diehl fire stated in full.
0.2 — DRAFT	October 2026	Tim Roxey	Conforms to the framework at v0.4 and takes in the return of Project 11's Defense Industrial Base batch and the author's rulings of 14 October 2026 to follow the criminal record and on state theft. The three reference events proposed at v0.1 are stated as conferred, as DBA-DIB-RE-1 (the Diehl fire), DBA-DIB-RE-2 (Alabuga) and DBA-DIB-RE-3 (the C-17 theft), at Sections 1.3, 5.1, 5.5, 6.2, 6.3, 9.2 and 10.6. Erratum 1 records the prosecutor's closure, describes the site as Diehl Metal Applications not supplying Diehl Defence, attributes the Journal's details, and moves the event from Section 5.2 to Section 5.1 under RECORD PRIMACY; Section 5.2 is left without a reference event, the question of another held for the author. The optional refinements add the Institute's imagery assessment of the 2024 strike and the C-17 dates and plea record. The six recommendations enter at Sections 3.3 and 4.4 (the index and the people in the holding), 5.2 (the security-assessment track), 5.8 and 6.3 (the workforce and its housing; protection designed against the campaign's weapon), 6.1 (applicability for every site carrying a defense group's name) and 6.9. Section 10.9 carries the notings to the coordination class and Defense Supply Chain. The cover lineage stays as issued. No change to the class's scope, chains, parameters, event categories or residuals; the v0.1 entry is left as a record.
0.3 — DRAFT	October 2026	Tim Roxey	On the author's decision of 14 October 2026 that Section 5.2, left without a reference event at v0.2, should seek a new one, proposes for it a fourth event, for the framework to confer: the fire and explosions at the Canadian Car and Foundry Company's shell plant at Kingsland, New Jersey, on 11 January 1917, found by the Mixed Claims Commission in 1930 not to have been caused by any German agent and in 1939, that finding set aside for fraud in Germany's evidence, to engage Germany's liability. Checked by an independent review agent, not a person, against the Commission's decision of 1930 and the Supreme Court's account;

Version	Date	Author	Description
			its corrections were taken in. Chosen by the author over the Vrbětice depot explosions (2014), a storage depot rather than a production plant, and the Leyton warehouse arson (2024), a commercial logistics site. Sections 5.2, 9.2, 10.6 and 10.9 and the references record it, with three points for the instrument at Section 10.6: the record is a tribunal's and parts within itself, the act was found in 1939 to engage a state's liability, in a then-neutral country, and no individual is named. No other change; earlier entries are left as records.